ChatGpt: Decreasing Universe Theory (DUT) —Summary

1. Central idea

The **Decreasing Universe Theory (DUT)** proposes that the Universe is **not actually expanding**. Instead, what changes over time are the **physical scales of matter** — particles, atoms, and characteristic lengths — which would **gradually decrease**.

Since all our measuring devices (rulers, clocks, spectroscopic standards) are also made of the same matter, the shrinking of physical scales would make the Universe **appear to expand** when comparing observations from different epochs.

Conceptually:

Universe appears to expand

Thus, observations usually interpreted as cosmic expansion would instead be interpreted as an **effect of changing physical scales**.

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2. Motivation of the theory

The DUT emerges as a conceptual alternative to cosmological models based on the **metric expansion of space**, such as the standard **ΛCDM model**.

Its motivations include:

- Searching for alternative explanations for the **cosmological redshift**.
- Avoiding the need for certain elements of the standard model, such as specific interpretations of **dark energy**.
- Exploring the possibility that **fundamental physical scales or constants evolve with cosmic time**.

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3. Explanation of redshift

In the standard cosmological model (ΛCDM), redshift is interpreted mainly as a consequence of the **expansion of space** between the emission and observation of light.

In the DUT, redshift would instead arise from changes in physical scales:

1. **Atoms in the past would be larger**, or would emit photons with different intrinsic properties.
2. Over time, the scale of matter would decrease.

3. Light emitted in the past would therefore be observed today with a **longer wavelength**, producing redshift.

Thus:

without requiring expansion of space itself.

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4. Theoretical implications

If the DUT were correct, several interpretations in cosmology would need to be reconsidered. Proposed implications include:

- **Cosmological redshift** reinterpreted as a scale-evolution effect.
- A possible alternative explanation for phenomena currently attributed to **dark energy**.
- A new interpretation of the relationship between **stellar age and observed redshift**.
- The possibility that some effects attributed to **dark matter** could instead be observational artifacts related to scale evolution or photon propagation effects.

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5. Comparison with the standard cosmological model (ΛCDM)

Aspect	**ΛCDM (standard model)**	**DUT**
Expansion of the Universe	Space expands	Physical scales of matter shrink
Redshift	Metric expansion of space	Evolution of physical scales or emission properties
Dark energy	Required to explain accelerated expansion	May not be necessary
Interpretation of cosmic evolution	Universe becomes larger over time	Physical scales become smaller over time

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6. Possible observational tests

For the DUT to be considered a viable alternative, it must reproduce the large body of existing cosmological observations. Important tests include:

- **Type Ia supernovae** (distance–redshift relations)
- **Cosmic Microwave Background (CMB)**
- **Baryon Acoustic Oscillations (BAO)**
- **Large-scale structure formation**
- **Relationships between galaxy properties and redshift**

Additional tests may involve comparing galaxies with:

- **the same mass**
- **the same distance measured independently of redshift**
- **different stellar ages**

to investigate correlations between **stellar population age and residual redshift**.

See also:

Some Articles about D.U.T:
https://jocaxx.github.io/DUniverse/

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the **Decreasing Universe Theory (DUT)** as it is presented in the available papers and summaries by João Carlos Holland de Barcellos.

Overview

The **Decreasing Universe Theory** proposes that the main large-scale cosmological effect is **not the expansion of space**, but a **continuous contraction of space and matter under gravity**. In this picture, the Universe appears to expand because the physical scales used to measure it are themselves shrinking over time. The theory is presented as an alternative to the standard ΛCDM model and claims to address cosmic redshift, accelerated expansion, dark matter effects, and some CMB-related questions without relying on the usual interpretation of expanding space, dark energy, or dark matter. (ResearchGate)

Core postulate

The central postulate is that the **gravitational field continuously contracts space and everything embedded in it**. That means not only astronomical structures but also local physical scales may shrink with time. In the theory's own framing, this contraction is

cumulative and universal, though stronger effects are often associated with stronger gravitational environments. (CME Live)

Conceptually, DUT replaces this standard idea:

with this alternative:

How DUT explains cosmological redshift

In standard cosmology, a distant galaxy's light is redshifted because space stretches while the photon travels. In DUT, the redshift is instead interpreted as an effect of **relative scale change between emission and observation**. The wavelength we measure today looks longer not because space expanded, but because the measuring framework and/or the emitting system changed scale over cosmic time. The theory's summaries explicitly present DUT as an alternative explanation for cosmological redshift. (CME Live)

In practical terms, DUT says the Universe may look as if distant galaxies are receding faster and faster, while the deeper mechanism is contraction of space/matter rather than expansion of space itself. (CME Live)

Claimed explanation of apparent accelerated expansion

One of the theory's recurring claims is that it can explain the **apparent accelerated expansion of galaxies without dark energy**. In the DUT framing, acceleration is not a genuine acceleration of the scale factor of the Universe; it is an observational consequence of contraction-driven scale changes. The theory therefore tries to replace the dark-energy interpretation with a contraction-based one. (CME Live)

Claimed explanation of dark matter effects

A second major claim is that DUT can reproduce galaxy rotation effects normally attributed to dark matter. One paper explicitly presents the "dark matter effect" as an apparent consequence of the DUT mechanism, arguing that contraction and photon-related effects can mimic the observed rotation-curve anomaly without requiring dark matter as a separate substance. (PhilPapers)

That is one of the theory's strongest advertised ambitions: a single mechanism that would reinterpret both

- apparent cosmic acceleration, and
- anomalous galactic dynamics. (CME Live)

Role of gravity in DUT

Gravity is not just an attractive force in the usual sense. In DUT, gravity is elevated into a more global geometric driver of contraction. The theory's descriptions say that

gravitational fields induce continuous spatial contraction, and that more massive systems should be associated with stronger contraction effects. (independent.academia.edu)

This point matters because it leads directly to one of DUT's main observational predictions.

Main observational prediction: mass vs. redshift at fixed distance

A repeated DUT prediction is the following:

For galaxies at the same distance, more massive galaxies should show **smaller redshifts** on average.

This is presented as the opposite of what the author argues the standard model would imply from ordinary gravitational redshift considerations. Several DUT papers frame this as a **falsifiable test**: compare galaxies at the same independently measured distance and see whether higher mass is associated with lower observed redshift. (PhilPapers)

That prediction is important because it turns DUT from a purely verbal reinterpretation into something testable against galaxy data.

Extension of the theory to stellar age

A related deduction, stated in summaries around the theory, is that a galaxy's **time under gravitational contraction**, meaning roughly its age or evolutionary maturity, could influence the observed redshift. In that interpretation, stellar age is not merely correlated with redshift through ordinary cosmic evolution; it becomes part of the DUT mechanism itself. This is why age–redshift tests are natural within the DUT framework. (independent.academia.edu)

DUT and the CMB

The theory has also been extended to the **Cosmic Microwave Background** and the **age of the Universe**. One paper explicitly says it uses DUT to estimate the age of the Universe from the average wavelength of the CMB radiation. That means DUT is not only trying to reinterpret galaxy redshift; it is also trying to offer an alternative account of one of the central pillars of modern cosmology. (PhilArchive)

Philosophical structure of the theory

At a conceptual level, DUT is a **scale reinterpretation theory**. Instead of saying "the cosmos gets bigger," it says "the standard of measurement gets smaller." This is why the theory often sounds counterintuitive: many observational signatures of expansion are preserved, but the ontological explanation is reversed.

So, in DUT:

- the Universe need not be expanding in the standard sense,
- redshift need not imply expanding space,
- dark energy may be unnecessary,
- dark matter may be unnecessary or partly illusory,
- gravitational contraction becomes the unifying mechanism. (CME Live)

What DUT would have to explain to become competitive

This is where the theory faces its real burden. A viable cosmological theory cannot explain only one or two effects. To compete with ΛCDM, DUT would need to reproduce, in a quantitatively consistent way:

- the Type Ia supernova distance–redshift relation,
- the detailed angular power spectrum of the CMB,
- baryon acoustic oscillations,
- large-scale structure formation,
- gravitational lensing observations,
- primordial nucleosynthesis constraints,
- local and high-redshift galaxy evolution data.

The publicly available DUT texts clearly attempt alternative explanations for some major effects, but from the standpoint of mainstream cosmology, the central question is whether the full observational package can be reproduced with comparable precision. The available sources present DUT as an alternative proposal, not as a theory that has already replaced ΛCDM in the literature. (ResearchGate)

Scientific status

As of the currently available sources, DUT appears as a **non-mainstream, author-driven alternative cosmological proposal**, with papers and summaries on repositories and journals outside the core consensus literature. The sources found include PhilArchive, ResearchGate summaries, vixra/MedCrave-style publications, and related postings. That means the theory exists in written form and makes explicit claims, but it is **not part of the standard accepted cosmological framework**. (PhilArchive)

Clean summary of DUT in one paragraph

The **Decreasing Universe Theory** says that the Universe may only **appear** to expand because gravity causes a long-term **contraction of space and matter**, changing the

physical scale of atoms, objects, and measurements over time. Under this interpretation, cosmological redshift is not primarily due to expanding space, apparent acceleration does not require dark energy, and some galaxy-dynamics anomalies may not require dark matter. The theory's signature empirical claim is that, at equal distance, more massive galaxies should tend to exhibit **smaller** redshifts, with possible related effects involving galaxy age. (PhilPapers)

Bottom line

DUT is best understood as a **global contraction cosmology**: a theory that tries to reinterpret the main observational evidence for expansion as evidence for shrinking physical scales instead. Its appeal is conceptual unification. Its weakness, at present, is the very high evidential bar required to replace ΛCDM.

If you want, I can turn this into a more formal essay-style exposition, with sections like **Foundations, Mechanism, Predictions, Comparison with ΛCDM, and Main Objections**.